PROJECT REPORT OF CO-OP PROJECT AT INDUSTRY (MODULE-2)

ON

DNS Benchmarking and System Optimization Tool

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

We hereby certify that the work which is being presented in the project report entitled **"DNS Benchmarking And System Optimization Tool"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of **Dr. Lalit Sharma.** The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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1. INTRODUCTION

Within modern situations, usage of technology makes internet efficiency very relevant for many processes. One of the key factors that affect internet speed and application efficiency is the Domain Name System (DNS). DNS stands for a map between domain names and IP addresses and is one of the main aspects of modern networking. DNS servers differ in efficiency, and it is crucial to consider the performance of DNS services.

With the growing dependence on cloud services and web-based services in general, it becomes necessary to find out which servers are more efficient in terms of performance. Very often, people simply forget about the possibility of choosing some different DNS servers instead of using services provided by their Internet providers. Public DNS services like Google DNS and Cloudflare DNS provide better and safer internet access compared to other servers. However, before making any choice, a detailed analysis of available options is needed.

In order to find out the best DNS server option for particular internet provider, an application comparing the performance of different servers was created. In addition to carrying out analysis, this tool allows exporting results as well as simulating certain actions related to DNS performance, such as DNS change.

Thus, the major goal of the current project is the creation of a program analyzing performance of DNS servers. With the help of GUI and network-related skills, this program will become informative for users.

2. PROBLEM STATEMENT

Today's networking environment is characterized by the experience of delay on the part of end-users who attempt to access various websites and online services. Despite the common belief that bandwidth is the major factor that affects speed, in reality, DNS is another critical issue since inefficient DNS can also cause the problem of latency and prevent users from establishing connections to the web server.

Many people lack both skills and the appropriate means to conduct proper DNS tests. Current solutions are complicated, ineffective, or offer no system-wide optimization options. This is why users continue using inefficient DNS settings.

What is needed is a universal system that would be able to:

- Conduct reliable benchmarks;
- Make comparative analyses of DNS performances;
- Offer the best DNS;
- Implement effective DNS settings.

3. SYSTEM REQUIREMENTS

3.1 Hardware Requirements

The hardware requirements to use the application are not high because the computational load is minimal while the actions the algorithm performs are rather simple. The users will not need any special equipment to use the application; however, the core attributes of the system may differ between common devices.

- **Processor**: Intel Core i3 and above (Intel Core i5 recommended)
- **RAM**: at least 4 GB (8 GB recommended)
- **Hard Disk**: available disk space should not be less than 200 MB
- **Internet Connection**: active to run DNS tests and benchmark them
- **Monitor**: standard computer monitor with resolution of 1366×768 or higher

All these features make it possible to efficiently use the application, including latency checking, image creation and exporting.

3.2 Software Requirements

Software requirements describe all the conditions that are necessary to run the application and work with all its capabilities, such as GUI generation, data processing and output.

- **Operating System**: Windows 10 or 11 /Linux /MacOS
- **Programming Language**: Python 3.x
- **Libraries/Modules:**
    - Tkinter (to generate the GUI)
    - Matplotlib (data visualization)
    - ReportLab (creation of PDF file for the test results)
    - OpenPyXL (.xlsx files processor)
- **IDE**: optional, but preferred (Visual Studio Code, for example)
- **Version Control System**: optional, Git may be used

All of these elements combine into the required application environment.

4. METHODOLOGY

The DNS Benchmarking and System Optimization Tool is designed with a well-thought-out methodology to enhance efficiency and effectiveness in terms of performance. The strategy encompasses understanding problems, developing systems, implementing features, testing, and ensuring usability, accuracy, and timeliness of performance.

4.1 Problem Analysis and Requirements

During the first step, there was a need to identify weaknesses in the current ways of measuring and establishing the most optimized DNS server. The discovery that many people use default DNS server settings but fail to assess their performance showed the need to develop a solution capable of measuring latencies and comparing different DNS servers. Therefore, some of the requirements identified included:

- Ability to test different DNS servers
- Accuracy in measuring latencies
- Presentation of test results using graphs or charts
- Exporting results into reports
- An easy-to-use graphical user interface (GUI)

4.2 System Design and Architecture

The proposed DNS benchmarking tool is designed as a desktop software whose architecture includes several modules that execute various processes within the system. Some of the modules in this architecture include:

- **User Interface Layer:** This is responsible for managing interaction between users and the computer program through forms, control buttons, data tables, and graphics.
- **Processing Layer:** This module measures latency of DNS servers using sockets.
- **Visualization Layer**: This shows results of the test using charts.
- **Export Module:** This helps in exporting results into various file formats such as PDF, CSV, and Excel.

4.3 DNS Benchmarking Implementation

DNS benchmarking is the most important feature in the design. Its implementation involves connecting to servers on port 53 and measuring time used. The process entails:

- Establishing connections to DNS servers using sockets from predefined lists and any newly created ones
- Measuring latencies of all DNS servers involved in the analysis
- Including timeout and error handling processes
- Storing results in structured form

4.4 GUI Design and User Interaction

For better usability of the developed program, GUI design becomes very critical. The GUI enables users to perform the following activities:

- Adding new DNS servers
- Initiating analysis process
- Viewing test results using graphs
- Changing of themes between light and dark modes

4.5 Visualization and Reporting of Results

Graphical methods were incorporated using charts to increase visual clarity and allow quick determination of the fastest DNS server. Moreover, reporting options were introduced to make it possible to output data in several formats:

- PDF to document the findings
- CSV to analyze them
- Excel to structure the obtained data

Thus, both visualization and exporting of the results can be done using this program.

4.6 Application Evaluation and Validation

Evaluation of this application took place in different conditions in order to ensure its effectiveness and reliability. Several DNS servers were tested to prove the accuracy of calculations performed by the program. During this process, such aspects were evaluated:

- Functionality of the program
- Responsiveness of the GUI
- Input of invalid arguments and handling of time-outs
- Exporting of results in different formats

Validation of data was performed to verify that correct comparisons are performed.

4.7 Deployment and Execution

The final step involved making this application ready for use by users. To achieve this goal, this program was created as an executable file to be able to run it without the Python environment. Issues taken into account during this stage include:

- Development of an executable file
- Compatibility with Windows ,Linux and Mac operating systems
- Minimal hardware requirements

5. SOFTWARE DESIGN

The software architecture for the DNS Benchmarking and System Optimization Tool focuses on providing simplicity, modularity, and efficiency of task completion. The program operates using a modular approach, meaning that each component of the program implements particular functions and processes. Such an approach is aimed at facilitating further software testing and ensuring its scalability.

5.1 Frontend (User Interface)

The frontend refers to an interface which allows users to interact with the program. It aims at presenting the data provided by the software in an informative way. Major components of the frontend are as follows:

- Fields allowing users to add custom DNS servers
- Buttons for initiating DNS analysis and generating output data
- Representation of results in a table format
- Representation of data graphically using charts
- Ability to switch between two themes – dark mode and light mode
- Log window showing progress
- Option to delete DNS according to need

The frontend is simple to use for people having little experience with computer networks and programming languages.

5.2 Backend (Processing and Control Logic)

The backend refers to the main part of the software responsible for implementing the required processing logic and serving as an intermediary between the interface and the underlying system components. Functions performed by this module include:

- Management of DNS server data (including both default and those added by users)
- Performing necessary measurement procedures
- Using multithreading to execute processes efficiently without affecting the interface
- Processing obtained results and preparing for their visualization

- Error handling and management (including such situations as timeout and unreachable servers)

5.3 DNS Benchmarking Engine

This module represents the key component of the application responsible for measuring performance characteristics of the DNS server. The process includes establishing connections to DNS servers through ports to obtain data on latency time.

Core functions of this module include:

- Sending messages to DNS server to establish a connection (port 53)
- Calculating latency time in milliseconds
- Identifying the best DNS server (by minimum latency time)
- Managing failed connections and annotating them appropriately

The benchmarking engine represents the primary analytical tool for measuring DNS performance.

5.4 Data Management and Reporting Module

This module handles the data storing, formatting and exporting procedures. After the generation of DNS results, data are processed in order to provide visualizable and presentable outputs. Tasks include:

- Storing results in structured manner (lists and tables)
- Exporting the data to file formats
- Generating reports in PDF, CSV and Excel formats
- Consistency of the data across different modules

This module improves tool's usability providing users with opportunities to save and distribute their analysis results.

5.5 Visualization Module

The visualization module takes responsibility for representing the data obtained in graphs. It is aimed at making the interpretation easier by replacing simple comparisons with visual ones. The visualization includes:

- Representation of DNS latency in bar graphs
- Determining the fastest DNS server
- Real time updates related to the analysis results
- Theme-based appearance ensuring consistency with user interface

This module improves tool's interactivity and user-friendliness.

5.6 System Optimization Support (DNS Configuration Guidance)

Apart from benchmarking functions, the tool provides users with the recommendations for system optimization, namely the identification of the most efficient DNS server. In case of users' decision, one can change the settings in DNS system accordingly. This module does the following:

- Finds out the fastest DNS server
- Performs DNS system simulation with new settings
- Provides a basis for automating the procedure in the future

5.7 Logging and Monitoring Module

In order to provide transparency and traceability of all processes, the application uses logging mechanism that saves each DNS testing step. This module covers:

- Logging DNS testing in real time
- Latency results and timeout display
- Automatic scroll down to the latest logs
- Distinguishing between separate analysis steps

6. IMPLEMENTATION DETAILS

6.1 Frameworks and Technologies

The DNS Benchmarking and System Optimization Tool relies on a range of effective and dependable technologies that support GUI software development, networking, data visualization, report creation, operating system communication, and other necessary functions. These technologies are selected to ensure the tool's efficiency and scalability.

- **Backend:** Main functions like measuring DNS latency, data manipulation, multithreading, and operating system interactions are coded in Python. Using native Python libraries such as socket, threading, and time helps in communicating with systems and assessing their performance.

- **Frontend:** The GUI is created with Tkinter, a library used for designing graphical interfaces. It allows for the creation of input fields, buttons, tables, and other interactive elements.

- **Data Visualization**: Matplotlib is used to visualize data, producing graphs like the bar chart that compares DNS server latencies.

- **Report Creation:** PDF reports are generated using ReportLab. Creating reports in Microsoft Excel format is supported by OpenPyXL. CSV reports can be created using built-in functionality in Python.

- **Operating System Interactions:** Platform-dependent modules like subprocess, os, and ctypes simulate configurations of DNS servers and perform other actions at the OS level.

- **Communication:** Network communication with DNS servers is conducted using socket programming, connecting to the DNS server on port 53.

6.2 Core Logic

Core logic of the application is created in order to ensure the most efficient yet accurate and sequential workflow which will result in an effective DNS benchmarking and provide smooth user interaction.

- **User Input:** The whole process starts when user starts the analysis or provides additional DNS servers through the interface. All defined and input DNS servers are combined together to ensure accuracy.
- **DNS Server Iteration:** All DNS servers are considered one after another in the process of preparation for testing. Every server is treated individually in order to provide precise measurement of latency.
- **Latency Measurement**: A connection to each server is made using sockets on port 53, and elapsed time is measured as a latency of particular server.
- **Data Collection and Storage:** Information obtained during testing is stored in a structure and saved for further use.
- **Result Processing:** Collected data are processed in order to choose the best DNS server and treat failed tests individually.
- **Visualization Generation:** Data obtained is passed to visualization component and converted into graphs. One of the charts used during presentation of results is bar chart.
- **Logging and Feedback:** All operations performed by the application are logged in real time, so user can get feedback information about current actions of program.
- **Result Presentation:** Results obtained by the application are presented visually and numerically in order to allow the user to see them clearly.
- **Export Options:** The user is provided with ability to export results in such formats as PDF, CSV, and Excel.

7. TESTING AND VALIDATION

Testing and validation form critical steps in the verification of the functionality, validity, and quality of the DNS Benchmarking and System Optimization Tool. As the project involves live network communications and user interactions, different types of testing had been performed to check the functionality and performance of the application.

7.1 Testing Methods Used

• Unit Testing

Unit tests have been carried out for separate components of the software to validate their proper operation in isolation. Among other things, the following units have been tested:

- DNS latency testing function
- Data management logic
- Export operations (PDF, CSV, Excel file creation)

These tests helped to find and fix possible issues during the early development stages.

• Integration Testing

In order to verify if the different modules work properly together, integration testing has been performed. It comprised the following:

- Communication between GUI and the background logic
- Data exchange between DNS tester and visualization modules
- Collaboration between analysis engine and the export features

The aim of such testing is to verify the seamless interaction between the different application modules.

• Functional Testing

Functional testing checks if an application meets its requirements in terms of functionality. The following features have been examined:

- Adding and analyzing DNS servers
- Showing latency results in table format

- Creating graphs and updating them
- Changing themes (dark and light)
- Exporting reports in various formats

All the mentioned features were tested to ensure that they are operational in any situation.

• User Testing

User testing is aimed at checking the application's usability and responsiveness in practice. In particular, the following points were considered during the interface testing:

- Navigation and usage convenience
- Correct displaying of results
- Feedback in logs and other messages
- DNS analyzing performance

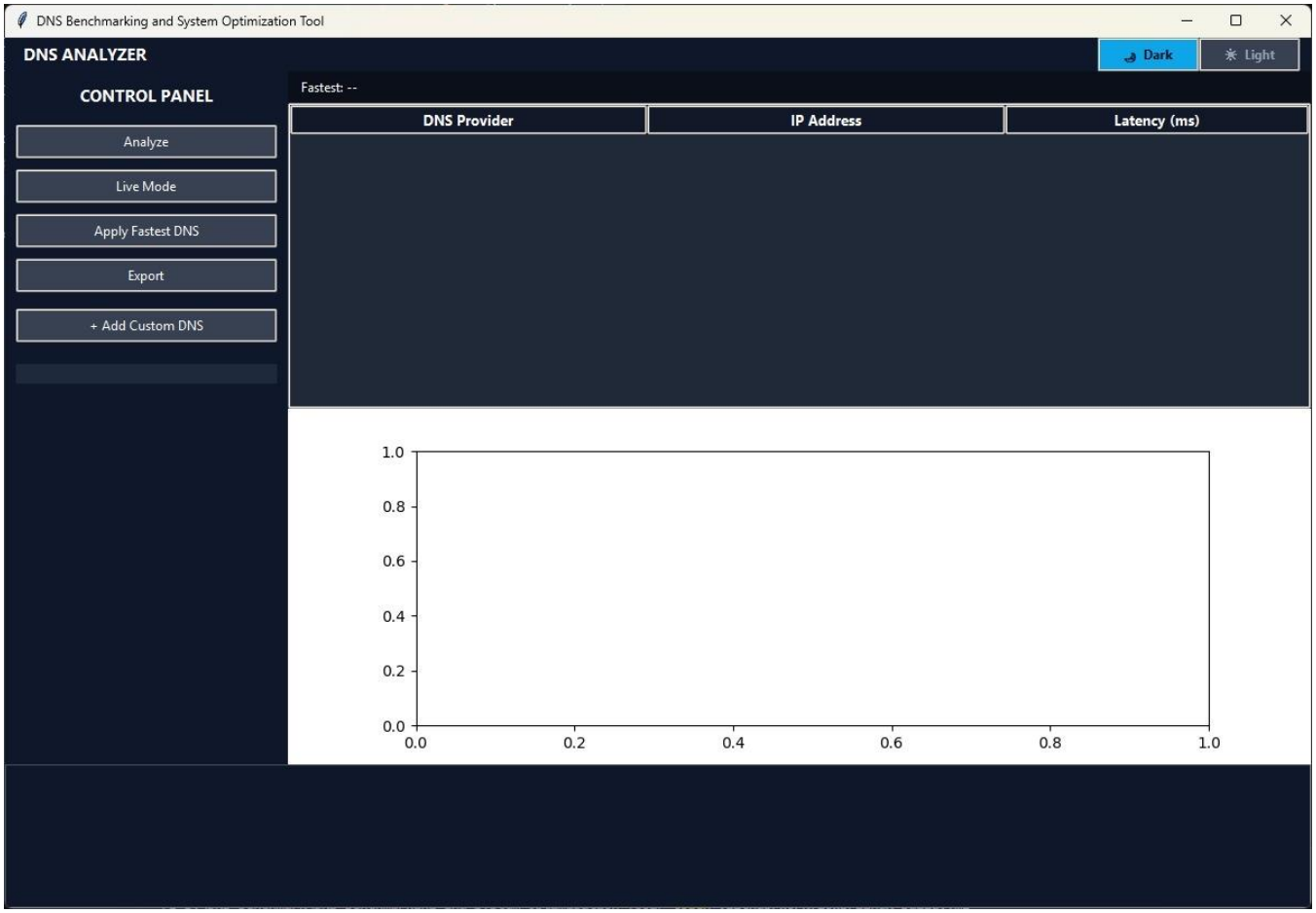
The testing allowed us to improve the user experience.

7.2 Validation of Results

For the purpose of validation, several tests have been performed to verify the correctness of DNS latency results provided by the application. During all the runs, the system managed to detect differences in response times and therefore proved its utility as DNS benchmarking tool. Various timeout situations and error-handling have been tested as well.

8. OUTCOMES

The outcomes achieved in the DNS Benchmarking and System Optimization Tool are based on practical experiments and observations concerning the performance of the DNS server under different conditions. In order to prove the reliability and accuracy of the tool, an appropriate experimental plan should be developed to verify it.

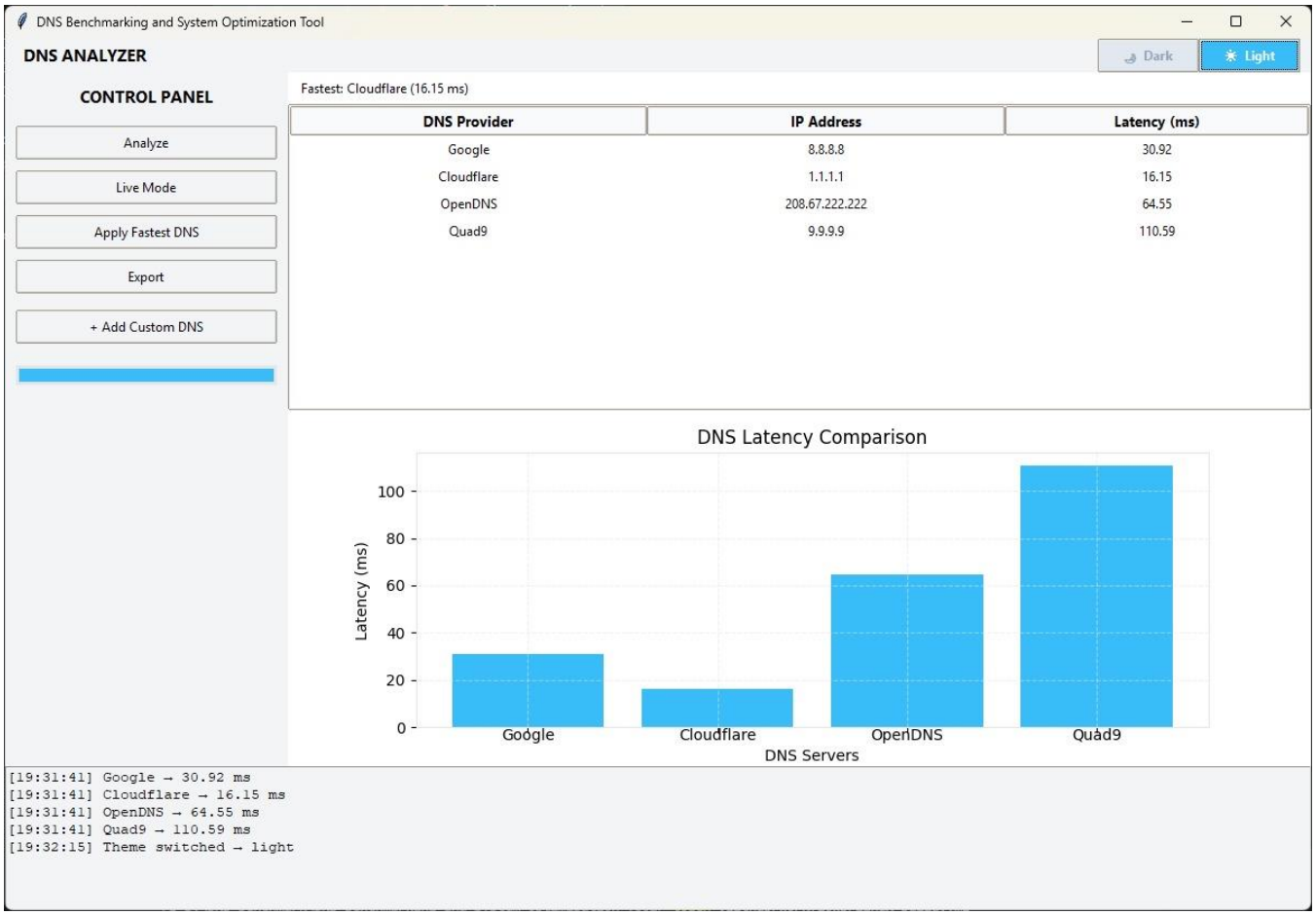


8.1 Experiment Plan to Receive Expected Outcome

To test the performance of the application, several experiments should be conducted, during which multiple DNS servers will be involved. In particular, both preconfigured and custom public DNS will be used to analyze their latency and choose the fastest one. To conduct such an experiment, the following steps were followed:

- Choose a number of DNS servers that are commonly used for testing purposes (Google, Cloudflare, OpenDNS, Quad9)
- Run the application multiple times in the same network environment
- Add a few custom DNS servers to see how flexible the tool is, delete if required
- Measure the latency for each DNS server
- Represent results in tables and graphs
- Export outputs in multiple formats

Such an experimental plan guarantees reliable results concerning the tool's performance.

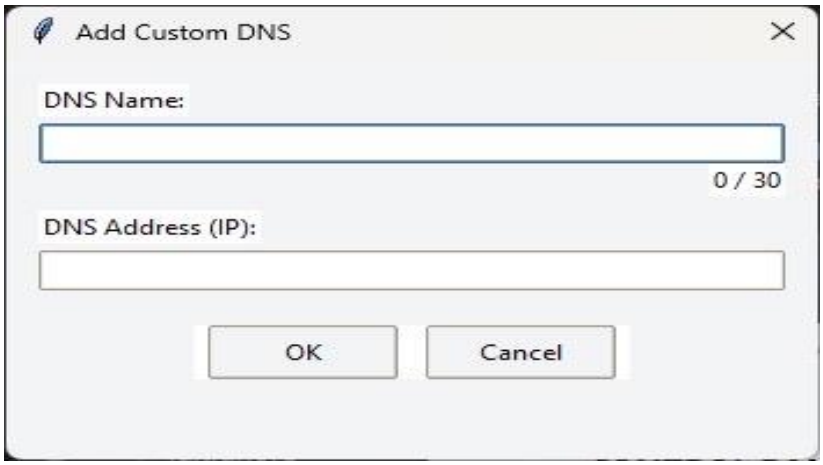


8.2 Outcomes

The results of the experiments conducted allowed confirming the effectiveness of the tool and its ability to deliver precise data regarding DNS performance.

- Latencies for all DNS servers were measured accurately
- The fastest DNS server was identified based on minimal latencies' values
- Differences between DNS performances were depicted in graphs correctly
- Timeouts and unreachable servers did not affect execution significantly
- Exports to other files (PDF, CSV, Excel) were completed correctly and structured adequately
- Application's UI worked well when analyzing data

In addition, the performance of the DNS might vary depending on network conditions; therefore, benchmarking is highly recommended before choosing a DNS server. All in all, the application met the expected outcome.



9. CONCLUSION

The concept of the DNS Benchmarking and System Optimization Tool was developed in order to cater to the demand for testing the effectiveness of DNS in a simple and efficient way. It was proven that network efficiency could be improved through choosing the optimal DNS server on the basis of analysis.

All of the primary goals set for the project were accomplished, namely measuring DNS latency, comparing several DNS servers, providing graph representation of the data and producing exportable reports. The tool appeared to be able to reliably detect any possible differences in performance between the tested DNS servers and provide their visual representation.

The importance of designing the user-friendly software has been proven since the developed tool is easy-to-use even for people lacking programming skills. Combining networking knowledge and user-friendly interface, the program can effectively serve as a bridge between them.

At the same time, this project creates an opportunity for implementing various improvements in the future such as the introduction of automatic DNS switch, additional analyses and more functions to enhance the performance of the application.

Thus, it can be concluded that this project was aimed at solving specific programming problems in accordance with the objectives stated previously.

10. REFERENCES

- Python Official Documentation
  https://docs.python.org/3/
- Matplotlib Documentation
  https://matplotlib.org/
- ReportLab Documentation
  https://docs.reportlab.com/reportlab/userguide/ch1_intro/
- Computer Networking Concepts (DNS)
  https://www.geeksforgeeks.org/computer-networks/domain-name-system-dns-in-application-layer/
- Google public DNS Documentation
  https://developers.google.com/speed/public-dns
- Cloudflare DNS Performance and Security Overview
  https://www.cloudflare.com
- OpenPyXl documentation
  https://openpyxl.readthedocs.io/en/stable/